2.6 Quiz: Lines, Exponents and Radicals — Markscheme

Version (a)

Total: 35 marks.

1. [Maximum mark: 10]

(a) $m_{AB} = \dfrac{5-1}{4-(-4)}$ **(M1)**

$m_{AB} = \dfrac{4}{8} = \dfrac{1}{2}$ **A1**

The M1 may be implied by the correct gradient: a bare answer earns (M1)A1. Wrong method: $\frac{\Delta x}{\Delta y} = 2$*: M0A0.*

(b) $y - 1 = \frac{1}{2}(x - (-4))$ **(M1)**

$y - 1 = \frac{1}{2}(x + 4)$ **A1**

M1 for substituting their gradient and either point; it may be implied by a correct equation, so a bare correct equation earns (M1)A1. Accept any correct form, e.g. $y = \frac{1}{2}x + 3$ *or* $x - 2y = -6$*; the answer must be an equation. FT from (a).*

(c) $m_L = -2$ **A1**

$y - 6 = -2(x - 1)$ **(M1)A1**

A1 for the negative reciprocal of their gradient. M1 for substituting their gradient and $C(1, 6)$*; it may be implied by a correct equation, so a bare correct equation earns A1(M1)A1. A1 for a correct equation in any form, e.g.* $y = -2x + 8$*. Wrong methods: the perpendicular bisector of* AB*,* $y = -2x + 3$*, passes through the midpoint* $(0,\ 3)$*, not through* C*: A1M0A0;* $m_L = 2$ *(no sign change) gives* $y = 2x + 4$*: A0M1A0. FT from (a).*

(d) $\frac{1}{2}(x + 4) + 1 = -2(x - 1) + 6$ **M1**

$\frac{1}{2}x + 3 = -2x + 8, \quad \frac{5}{2}x = 5$

$x = 2, \quad y = 4 \qquad (2,\ 4)$ **A1A1**

M1 for solving their two lines together by any method (equating, substitution, GDC). A1 for each coordinate. Wrong method: the midpoint of AB*,* $(0,\ 3)$*: M0A0A0. FT from (b) and (c).*

2. [Maximum mark: 7]

(a) $2x + 3y = 47$ **A1**

$x + 4y = 41$ **A1**

(b) $x = 41 - 4y, \quad 2(41 - 4y) + 3y = 47$ **(M1)**

$5y = 35$

adult ticket $x = 13$ euros **A1**

child ticket $y = 7$ euros **A1**

M1 for any valid method, including the GDC simultaneous-equation solver; a correct GDC answer with no other working earns full marks. FT from (a).

(c) $2(13) + 7k = 61$ **M1**

$7k = 35, \quad k = 5$ children **A1**

Wrong method: swapping the prices, $2(7) + 13k = 61$*, gives* $k = 3.62$*, not a whole number: M0A0. FT from (b).*

3. [Maximum mark: 10]

(a) $(2a^3b)^2 \times 3a^{-2}b^{-4} = 4a^6b^2 \times 3a^{-2}b^{-4}$

$= \dfrac{12a^4}{b^2}$ **A1A1A1**

Marks are for the final answer, not the steps: A1 for the coefficient 12*, A1 for* a^4*, A1 for* b^2 *in the denominator, independently.* $12a^4b^{-2}$ *earns A1A1A0 (the question asks for positive exponents only). Wrong method: leaving the* 2 *unsquared gives* $\frac{6a^4}{b^2}$*: A0A1A1.*

(b) $\left(\dfrac{8}{27}\right)^{-\frac{2}{3}} = \left(\dfrac{27}{8}\right)^{\frac{2}{3}} = \left(\dfrac{3}{2}\right)^2$ **(M1)**

$= \dfrac{9}{4}$ **A1**

M1 for inverting the fraction for the negative exponent, or for converting to cube-root form ($\sqrt[3]{8}$*,* $\sqrt[3]{27}$ *or* $\frac{2}{3}$*,* $\frac{3}{2}$ *seen); it may be implied by the correct answer, so a bare* $\frac{9}{4}$ *earns (M1)A1, with a feedback note to show the working. Accept* $2\frac{1}{4}$ *or* 2.25*. Wrong method: ignoring the negative sign gives* $\frac{4}{9}$*: M0A0.*

(c) $(3^2)^{x-1} = (3^3)^{x-3}$ **(M1)**

$2x - 2 = 3x - 9$ **A1**

$x = 7$ **A1**

M1 for writing both sides as powers of 3*; it may be implied by a correct* $2x - 2 = 3x - 9$*. A bare* $x = 7$ *earns M0A0A1. Wrong method:* $2x - 2 = 3x - 3$ *(the* 3 *not multiplied through) gives* $x = 1$*: M1A0A0.*

(d) $2^{x+3} = 2^{-5}$ **(M1)**

$x + 3 = -5, \quad x = -8$ **A1**

M1 for -5 *seen anywhere, e.g.* $\frac{1}{32} = 2^{-5}$ *or* $x + 3 = -5$*; it may be implied by the correct answer, so a bare* $x = -8$ *earns (M1)A1. Wrong method:* $\frac{1}{32} = 2^5$ *gives* $x = 2$*: M0A0.*

4. [Maximum mark: 8]

(a) $5\sqrt{2} + 2\sqrt{2} - 4\sqrt{2}$ **(M1)**

$= 3\sqrt{2}$ **A1**

M1 for writing at least two of the surds as multiples of $\sqrt{2}$; it may be implied by the correct answer, so a bare $3\sqrt{2}$ earns (M1)A1, with a feedback note to show the working. Wrong method: $\sqrt{50 + 8 - 32} = \sqrt{26}$: M0A0.

(b) $9 - 3\sqrt{5} + 3\sqrt{5} - 5$ **(M1)**

$= 4$ **A1**

Accept $3^2 - \left(\sqrt{5}\right)^2$ for the M1. The M1 may be implied by the correct answer: a bare 4 earns (M1)A1, with a feedback note to show the working. Wrong method: $9 + 5 = 14$: M0A0.

(c) $\dfrac{8}{3 - \sqrt{5}} \times \dfrac{3 + \sqrt{5}}{3 + \sqrt{5}} = \dfrac{8\left(3 + \sqrt{5}\right)}{4}$

$= 6 + 2\sqrt{5}$ **A1A1**

No method mark. A1 for $a = 6$, A1 for $b = 2$, independently. A bare $6 + 2\sqrt{5}$ earns A1A1, with a feedback note to show the working. Accept $a = 6$, $b = 2$ written separately for both marks, with a feedback note that the question asks for the form $a + b\sqrt{5}$. FT from (b).

(d) $4 = 3x\sqrt{2} - x\sqrt{2} = 2x\sqrt{2}$ **(M1)**

$x = \dfrac{4}{2\sqrt{2}} = \dfrac{2}{\sqrt{2}} = \sqrt{2}$ **A1**

M1 for collecting the x-terms on one side; it may be implied by the correct answer, so a bare $x = \sqrt{2}$ earns (M1)A1, with a feedback note to show the working. The A1 needs $\sqrt{2}$: $\frac{2}{\sqrt{2}}$ or 1.41 earns A0.

Version (b)

Total: 38 marks.

1. [Maximum mark: 12]

(a) $4y = -3x + 24, \quad y = -\frac{3}{4}x + 6$ **(M1)**

$m_1 = -\dfrac{3}{4}$ **A1**

M1 for rearranging to $y = mx + c$, or for using $-\frac{a}{b}$; it may be implied by the correct gradient, so a bare answer earns (M1)A1.

(b) $m_2 = \dfrac{4}{3}$ **A1**

$y - (-3) = \frac{4}{3}(x - (-1))$ **(M1)**

$y + 3 = \frac{4}{3}(x + 1)$ **A1**

M1 for substituting their gradient and $P(-1, -3)$; it may be implied by a correct equation, so a bare correct equation earns A1(M1)A1. Accept $y = \frac{4}{3}x - \frac{5}{3}$ or $4x - 3y = 5$. Wrong methods: $m_2 = -\frac{4}{3}$ (no sign change): A0M1A0; the perpendicular bisector of AB, $y = \frac{4}{3}x - \frac{7}{3}$, passes through the midpoint $(4,\ 3)$, not through P: A1M0A0. FT from (a).

(c) $3x + 4\left(\frac{4}{3}(x + 1) - 3\right) = 24$ **M1**

$\frac{25}{3}x = \frac{92}{3}, \quad x = \dfrac{92}{25} = 3.68$ **A1**

$y = \dfrac{81}{25} = 3.24$ **A1**

M1 for solving the two equations together by any method, including the GDC. Accept fractions or the exact decimals. Wrong method: the midpoint of AB, $(4,\ 3)$: M0A0A0. FT from (b).

(d) $A(8,\ 0)$, $B(0,\ 6)$, $\quad AB = \sqrt{8^2 + 6^2} = 10$ **A1**

$PF = \sqrt{(3.68 + 1)^2 + (3.24 + 3)^2}$ **M1**

$PF = \sqrt{60.84} = 7.8$ **A1**

Area $= \frac{1}{2}(10)(7.8) = 39$ **A1**

A1 for $AB = 10$, or for both intercepts if another method is used. M1 for any valid area method that uses P: base AB with height PF, an enclosing rectangle ($81 - 13.5 - 24 - 4.5$), or the shoelace formula. A1 for the height 7.8, or the correct intermediate of their method. Wrong methods: the area of triangle OAB, 24; $\frac{1}{2}(AB)(PA)$ with $PA = 3\sqrt{10}$: M0. FT from (c).

2. [Maximum mark: 7]

Label the equations (1), (2), (3).

$3 \times (1) + 4 \times (2)$: $\quad 6x + 9y - 12z + 20x - 8y + 12z = -60 + 136$ **M1**

$26x + y = 76$ **A1**

$(1) + 2 \times (3)$: $\quad 2x + 3y - 4z + 6x + 8y + 4z = -20 + 22$ **M1**

$8x + 11y = 2$ **A1**

$y = 76 - 26x, \quad 8x + 11(76 - 26x) = 2, \quad -278x = -834$

$x = 3$ **A1**

$y = 76 - 26(3) = -2$ **A1**

$3(3) + 4(-2) + 2z = 11, \quad z = 5$ **A1**

Any order of elimination is acceptable. First M1 for scaling and combining two equations so that one variable cancels; A1 for the correct two-variable equation. Second M1 for eliminating the same variable from a different pair; A1 for the correct second equation. Then A1 for each of $x = 3$, $y = -2$, $z = 5$, independently; the final values may come from any valid method for the two-variable system. FT within the problem: a student who makes one arithmetic slip in an elimination loses that A1 and may earn the later A1s for values that follow correctly from their two equations. Wrong methods: adding or subtracting equations without scaling, so no variable cancels: M0; a sign error when subtracting (e.g. $-8y$ treated as $+8y$): M1A0. Answers from the GDC solver with no elimination shown earn M0A0M0A0A1A1A1 (the question asks for elimination). Substitution is not elimination: solving one equation for a variable and substituting it into the others earns no credit (0/7), even when the final values are correct.

3. [Maximum mark: 11]

(a) $(5^2)^{x+1} = (5^3)^{2x-3}$ **(M1)**

$2x + 2 = 6x - 9$ **A1**

$x = \dfrac{11}{4}$ **A1**

Accept 2.75. *M1 for writing both sides as powers of* 5 *correctly (*$25 = 5^2$, $125 = 5^3$*); it may be implied by a correct equation in the exponents. A wrong power of* 5 *(e.g.* $125 = 5^4$*) earns M0A0A0. Accept* $2(x + 1) = 3(2x - 3)$ *for the first A1. A bare* $x = \frac{11}{4}$ *earns M0A0A1. Wrong methods:* $2x + 1 = 6x - 9$ *or* $2x + 2 = 6x - 3$ *(a bracket not multiplied out) gives* $\frac{5}{2}$ *or* $\frac{5}{4}$*: M1A0A0.*

(b) $2a + 2b\sqrt{3} + a\sqrt{3} + 3b = 12 + 7\sqrt{3}$ **M1**

$2a + 3b = 12, \quad a + 2b = 7$ **A1**

$a = 3, \quad b = 2$ **A1**

Accept the conjugate route: $\frac{12+7\sqrt{3}}{2+\sqrt{3}} \times \frac{2-\sqrt{3}}{2-\sqrt{3}} = \frac{24-12\sqrt{3}+14\sqrt{3}-21}{1} = 3 + 2\sqrt{3}$ *(M1A1A1). Accept trial with the expansion of* $(2 + \sqrt{3})(3 + 2\sqrt{3})$ *shown as a check (M1A1A1). A bare answer earns M0A0A1.*

(c) $9^x = (3^x)^2 = u^2$, so $3u^2 - 82u + 27 = 0$ **M1**

$(3u - 1)(u - 27) = 0$ **(M1)**

$u = \dfrac{1}{3}$ or $u = 27$ **A1**

$3^x = \dfrac{1}{3} \Rightarrow x = -1$ **A1**

$3^x = 27 \Rightarrow x = 3$ **A1**

M1 for the quadratic in u*. (M1) for an attempt at any method of solving their quadratic (factorizing, the formula, completing the square or the GDC); it may be implied by a correct value of* u*. A1 for either* $u = \frac{1}{3}$ *or* $u = 27$ *correct. Stopping at* $u = \frac{1}{3}, 27$ *earns M1(M1)A1A0A0. Wrong method: reading* 3^x *as* $3x$ *gives* $x = \frac{1}{9}, 9$*: A0A0 for those values.*

4. [Maximum mark: 8]

(a) $u_n = 7 + 3(n-1)$ **A1**

$v_n = 2(1.5)^{n-1}$ **A1**

Accept $u_n = 3n + 4$.

(b) $2(1.5)^{n-1} > 7 + 3(n-1)$ **(M1)**

$u_7 = 25$, $v_7 = 22.8$; $\quad u_8 = 28$, $v_8 = 34.2$ **A1**

$n = 8$ **A1**

M1 for the inequality, or a GDC table or graph of both sequences; it may be implied by the values on both sides of the change. A1 for the values on both sides of the change. A bare $n = 8$ *earns M0A0A1. FT from (a):* $v_n = 2(1.5)^n$ *gives* $n = 7$.

(c) $\dfrac{2\left(1.5^n - 1\right)}{1.5 - 1} > \dfrac{n}{2}\left(2(7) + 3(n-1)\right)$ **M1**

$n = 9$: $149.8 < 171$; $\quad n = 10$: $226.7 > 205$ **A1**

$n = 10$ **A1**

M1 for both sum formulas (or a GDC table of both sums). A1 for the sums on both sides of the change. Wrong method: comparing terms instead of sums gives $n = 8$*: M0A0A0. FT from (a).*